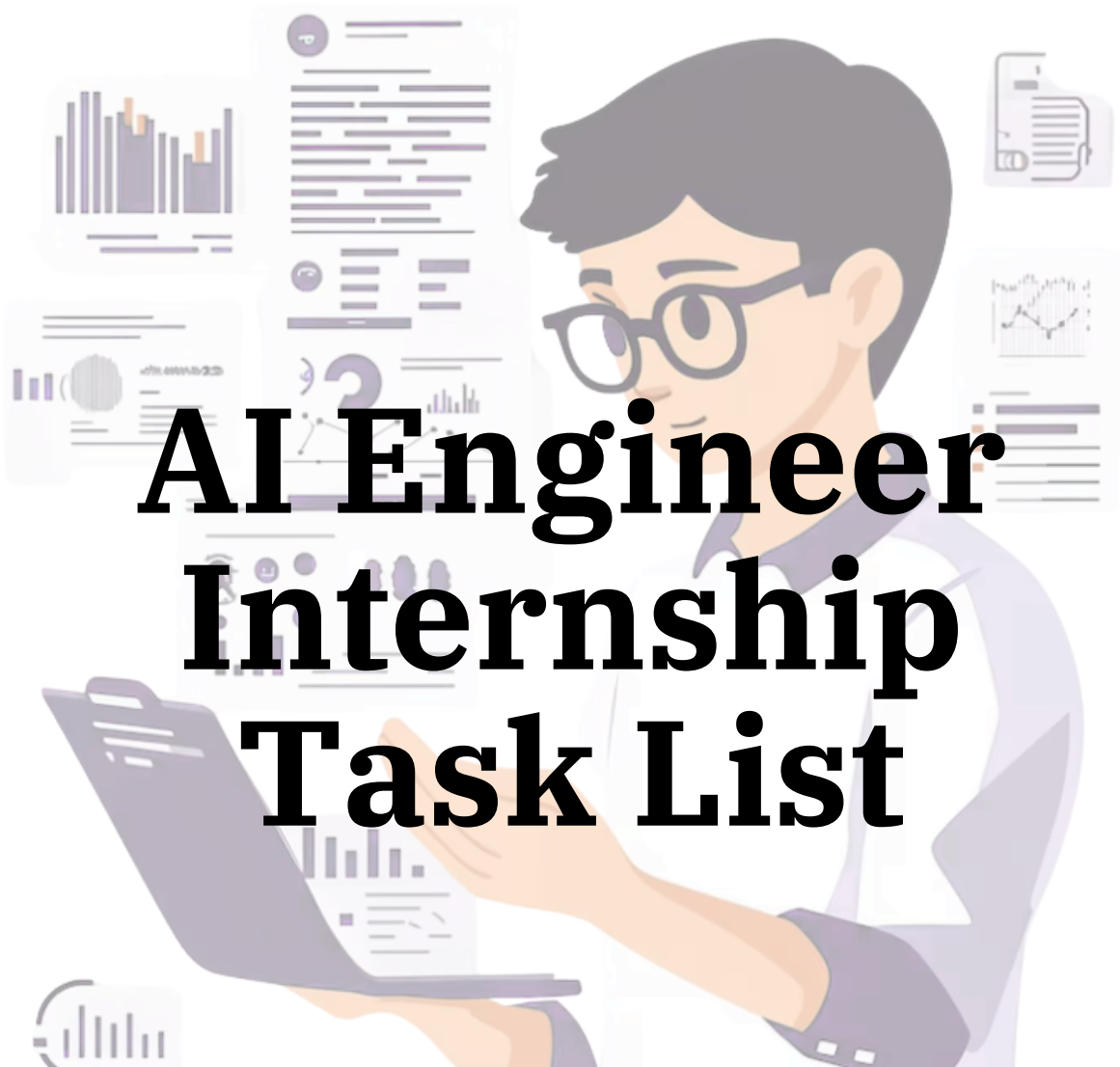




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A stylized illustration of a person with dark hair and glasses, wearing a light blue shirt, sitting at a desk and working on a laptop. The background is filled with various data visualization elements, including bar charts, line graphs, and document icons, all in a muted purple and blue color scheme. The overall style is clean and professional.

# **AI Engineer Internship Task List**



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## **Role: AI Engineer**

This internship track is designed to evaluate candidates progressively across beginner, intermediate, and advanced levels. The goal is to assess not only technical execution, but also requirement understanding, applied AI workflow design, product thinking, implementation clarity, and readiness to contribute to practical AI engineering tasks.

The last page contains the submission process, kindly read that page carefully.

### **Beginner Level:**

At the beginner level, you will be asked to build an AI-powered student utility application using an LLM API.

The project should be designed as a simple but useful tool that helps students with tasks such as note summarisation, quiz generation, answer improvement, concept explanation, or study assistance through prompt-based AI interaction.

The student utility app should typically include:

- A simple and usable interface
- A text input or content input area
- At least one meaningful AI-powered utility feature
- Prompt-based output generation
- Basic validation for empty or invalid inputs
- Error handling for failed API responses
- Clear display of generated output
- A clean and understandable user flow

You will be expected to integrate an AI model API properly and build a usable application around it instead of simply making isolated prompt calls. The app should solve a small but realistic student use case and should demonstrate that the candidate understands how to convert a practical idea into a working AI-powered feature.



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## **Role: AI Engineer**

The final application should allow the user to input content or a question, send that content to the model using a structured prompt, and display the generated output in a readable format. You may include features such as note summarisation, short quiz generation from notes, rewriting weak answers into better ones, or generating study explanations for selected topics.

You should also apply basic prompt design thinking so that outputs are relevant, understandable, and aligned with the intended use case. Input validation, prompt clarity, and usability of the output should all be considered as part of the solution.

You will be expected to explain how the app works, what prompt structure was used, how errors were handled, and how the application provides a useful AI-assisted student workflow rather than functioning as a generic chatbot only.

## **Purpose of this level**

This stage is meant to evaluate beginner-level AI application development fundamentals such as API integration, prompt-based thinking, simple UI or app flow design, basic validation, output handling, and the ability to turn a small real-world problem into a usable AI-powered solution.



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## Intermediate Level:

At the intermediate level, you will be asked to build a document-based question answering assistant using embeddings and retrieval over uploaded text or PDF content.

The project should be developed as a practical AI workflow where the user can upload one or more documents, ask questions based on those documents, and receive answers grounded in the uploaded content rather than generic model responses.

The document-based Q&A assistant should typically include:

- File upload support for text or PDF documents
- Document text extraction
- Chunking of long content into smaller sections
- Embedding generation for document chunks
- Vector search or similarity-based retrieval
- Question answering grounded in retrieved context
- Display of retrieved source context or supporting text
- A usable interface for upload and querying
- Basic handling for invalid files or empty queries

You will be expected to move beyond basic API calling and implement a proper retrieval-based workflow. This means the system should not answer questions from model memory alone, but should first identify relevant content from the uploaded documents and then use that context to generate a grounded response.

The final application should show a practical understanding of how document ingestion, chunking, embeddings, retrieval, and answer generation work together in a useful AI pipeline. The interface does not need to be overly complex, but it should clearly support document upload, question input, answer display, and context visibility.

You are encouraged to demonstrate how you selected chunk sizes, how document content is processed, how relevant sections are retrieved, and how the final answer remains tied to the document instead of drifting into unsupported outputs.



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## Role: AI Engineer

You should also be able to explain how the retrieval flow was structured, what embedding or vector search approach was used, how grounding was handled, and how the system reduces the chances of irrelevant or hallucinated responses.

### Purpose of this level

This stage is meant to evaluate practical AI engineering workflow understanding, including document ingestion, embeddings, chunking strategy, vector retrieval, grounded generation, application structuring, and the ability to simulate a realistic AI-assisted knowledge workflow rather than only a prompt-based feature.





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## Advanced Level:

At the advanced level, you will be asked to build a production-style Retrieval-Augmented Generation assistant using tools such as FastAPI, Streamlit, Pydantic, LangGraph, FAISS, Docker, and any suitable AI model for grounded Q&A over PDF, TXT, and Markdown files.

The project should be designed as a more complete AI engineering system rather than a simple prototype. It should reflect how modern AI applications are structured with multiple components, better validation, clearer architecture, and more robust handling of document-based user interaction.

The production-style RAG assistant should typically include:

- Multi-format document ingestion
- Parsing and preprocessing
- Chunking strategy
- Embedding generation
- Vector indexing and search
- Document-scoped retrieval
- Grounded answer generation
- Context display or citation support
- Query rewriting or refinement where useful
- Reranking or retrieval improvement logic
- Validation using typed schemas
- Streaming or responsive answer delivery
- Clean API or app structure
- Containerized setup for reproducibility

You will be expected to design an agentic or multi-step workflow that reflects practical AI engineering thinking. This may include ingestion, chunking, embedding generation, retrieval, reranking, groundedness checks, response generation, and final response delivery through a clean interface or API.

The final system should not only answer questions, but should also show signs of production-style thinking such as clear architecture, modular components, reliable input handling, well-structured request and response models, and separation between ingestion, retrieval, and generation responsibilities.

You should aim to demonstrate how AI quality is improved through retrieval orchestration, prompt control, context filtering, and validation checks. You may also include observability-oriented features such as retrieved chunk display, confidence-style indicators, or structured responses to help show how the system behaves internally.



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## Advanced Level:

You will be expected to explain the architecture of the full system, why each component was used, how retrieval quality was handled, how hallucinations were reduced, and how the project reflects practical AI engineering beyond a basic demo application.

### Purpose of this level

This stage is meant to evaluate advanced AI engineering capability, including system design, modular workflow construction, grounded retrieval pipelines, typed backend structure, production-style tooling, reliability thinking, and the ability to build a more realistic end-to-end AI application that resembles modern LLM product development work.

### Submission Process:

- If you have any doubts related to the task, you may contact the assigned mentor through the relevant email address at [g2doubts@shadowfox.in](mailto:g2doubts@shadowfox.in).
- You may also use the above email address for task validation to ensure that you are on the right track.
- For the final submission, you are required to submit the GitHub repository containing all the tasks. Additionally, it is mandatory to submit a 3 to 5 minute self-recorded video explaining your work. The video must be shared through a Google Drive link in the submission form.

**Note: Mailing the mentor would not guarantee the issuance of a certificate. Details regarding the final submission process for the completion certificate will be shared separately in your respective WhatsApp group.**